Friction: Test the Surfaces · TA Brief

Learning goal: I can compare friction on different run-out surfaces using repeated measurements and a controlled setup.

Watch (staff preview first)

Short silent teacher demonstration clip for this lesson's model and practical sequence · preview before class use.

Adult Support Routine

No cold calling; nobody is asked to read difficult text aloud.

Give time to start -> point to the relevant place -> rephrase -> offer a small prompt or word bank -> reduce to a limited choice.

Beside the You do in the pupil book, mark I (independent), S (supported) or C (choice).

Must-Know Words

run-out surface: the flat test surface at the bottom of the ramp where the block slows and stops

stopping distance: how far the block travels along the run-out surface before stopping, in cm

control variable: a factor kept the same so the surface comparison is fair

repeat trial: running the same surface test more than once to check consistency

Key Answers

A. To test the surface, only the run-out surface should change while block, ramp, and release stay controlled.

A. Keeping the block, ramp height, and release identical isolates the effect of the run-out surface.

1. 24 cm (shorter stopping distance). 2. Same block, same ramp height, same release point, no push. 3. It is part of the evidence; we can flag uncertainty, check the method, and repeat honestly.